Software Requirements Engineering
Assignment 3

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Case Study: Chemical Tracking Information System (CTIS) — Contoso Pharmaceuticals

1. Use-Case Diagram

The diagram below represents all major actors and use cases for the CTIS, including Chemists, Stockroom Staff, H&S; Officers, Admin/IT, and Government Regulators.

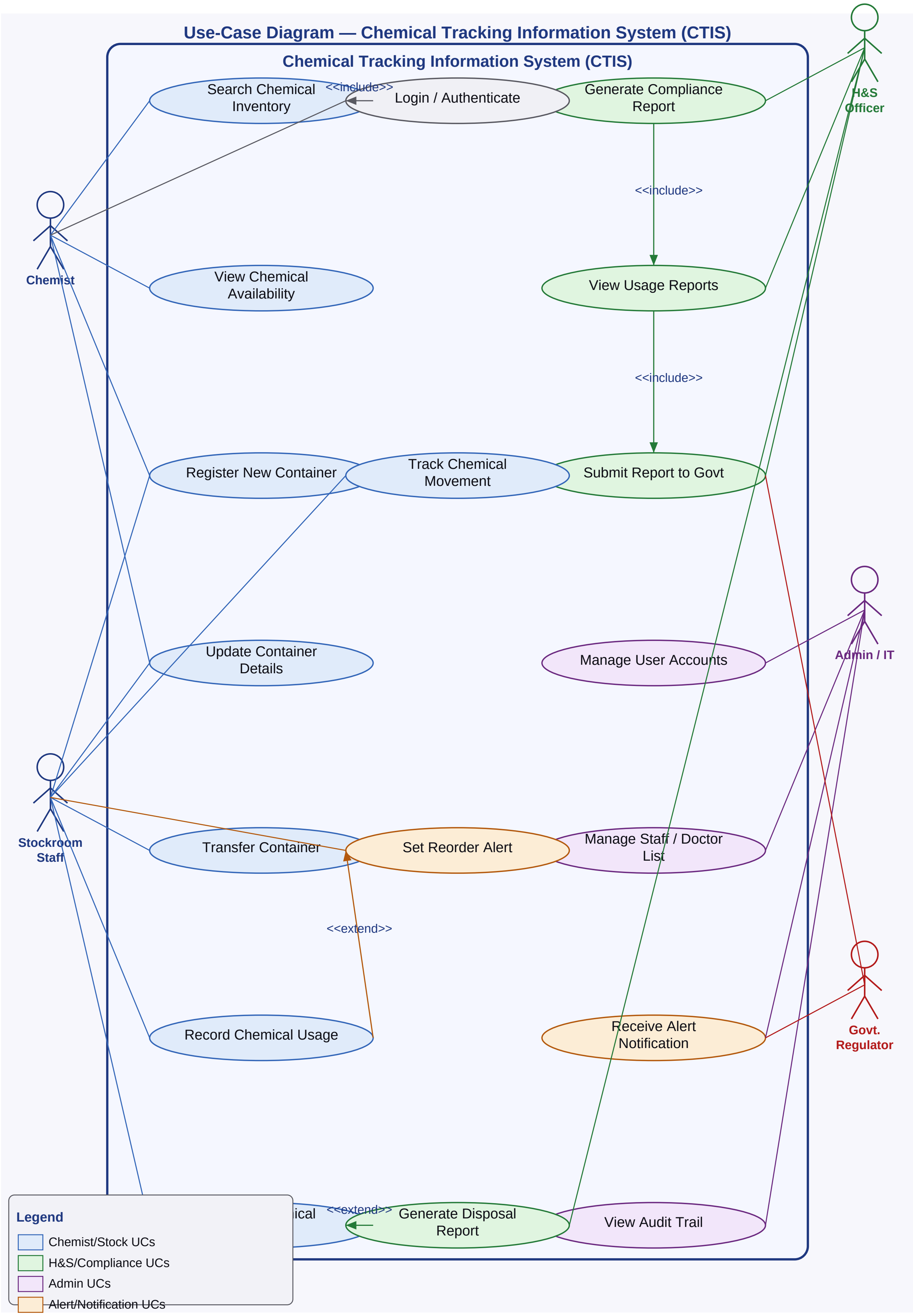

Figure 1: CTIS Use-Case Diagram

2. Fully-Dressed Use Cases

The following four use cases are described in fully-dressed format, covering normal flow, alternative flows, and exception flows.

Use Case 1: Search Chemical Inventory

Use Case ID	UC-01
Use Case Name	Search Chemical Inventory
Actor(s)	Chemist (Primary), Stockroom Staff (Secondary)
Description	A chemist searches the CTIS to find whether a required chemical is already available in any stockroom or laboratory before placing a new purchase order.
Preconditions	1. The user is authenticated and logged into CTIS. 2. Chemical inventory database is populated.
Postconditions	1. The user views a list of matching chemical containers with location and quantity. 2. If requested, a transfer is initiated.
Trigger	The chemist needs a specific chemical and checks CTIS before ordering.
Normal Flow	1. Chemist selects 'Search Chemicals' from the main menu. 2. System displays the search interface. 3. Chemist enters search criteria (chemical name, CAS number, location, or quantity range). 4. System validates the input format. 5. System queries the inventory database. 6. System displays matching containers with name, location, quantity, and container ID. 7. Chemist selects a container to view full details. 8. Chemist decides to request a transfer or exits the search.
Alternative Flow	A1 – Request Transfer: At step 8 chemist clicks 'Request Transfer'; system launches UC-05 (Transfer Container). A2 – Refine Search: At step 6 chemist modifies criteria and re-searches.
Exception Flow	E1 – Invalid Input (step 4): System highlights invalid field; returns to step 3. E2 – No Results Found: System displays 'No chemicals matching your criteria'; offers redirect to procurement request form; use case ends.
Business Rules	- Search available to all authenticated users. - Results show real-time quantity; reserved containers flagged. - Chemical names searchable via partial strings.
Frequency	Multiple times daily.
Priority	High

Use Case 2: Register New Chemical Container

Use Case ID	UC-02
Use Case Name	Register New Chemical Container
Actor(s)	Stockroom Staff (Primary), System (Secondary)

Description	Stockroom staff registers a newly received chemical container into the CTIS inventory so it becomes visible and searchable to all users.
Preconditions	1. Staff authenticated with Stockroom Staff or higher role. 2. Physical container on hand with label information.
Postconditions	1. Container record saved with unique ID. 2. Audit trail entry created. 3. H&S; notified if chemical is hazardous.
Trigger	A new chemical shipment arrives at the stockroom.
Normal Flow	1. Staff selects 'Register New Container'. 2. System presents the registration form. 3. Staff enters: chemical name, CAS number, quantity, unit, supplier, expiry date, storage location, and hazard classification. 4. System checks for duplicate containers. 5. System validates all mandatory fields. 6. System assigns a unique Container ID and timestamp. 7. System saves the record to the database. 8. System logs the action in the audit trail. 9. If hazardous, system sends notification to H&S; Officer. 10. System displays success confirmation with Container ID.
Alternative Flow	A1 – Batch Registration: Staff uploads a CSV file; system validates each row and registers valid entries; reports failed rows for correction.
Exception Flow	E1 – Duplicate Detected (step 4): System warns of existing container; staff confirms or cancels. E2 – Validation Failure (step 5): System highlights invalid fields; returns to step 3. E3 – Database Error (step 7): System prompts retry; entry not committed.
Business Rules	- CAS + supplier + batch = uniqueness key. - Hazardous chemicals (GHS 1–5) must trigger H&S; notification. - Expiry date cannot be in the past.
Frequency	Daily.
Priority	High

Use Case 3: Record Chemical Disposal

Use Case ID	UC-03
Use Case Name	Record Chemical Disposal
Actor(s)	Stockroom/Lab Staff (Primary), H&S; Officer (Secondary), System (Tertiary)
Description	Staff records the disposal of a chemical container so that disposal data is captured for compliance reporting and inventory is updated.
Preconditions	1. User is authenticated. 2. Container exists in CTIS with status 'Active'.
Postconditions	1. Container status updated to 'Disposed'. 2. Disposal record added to compliance log. 3. H&S; Officer notified if hazardous.
Trigger	A container is empty, expired, or requires disposal per safety guidelines.
Normal Flow	1. Staff selects 'Record Disposal'. 2. Staff enters or scans the Container ID. 3. System retrieves and displays container details. 4. Staff enters disposal details: date, method, quantity disposed, and reason. 5. System validates the disposal method against the approved list. 6. System updates container status to 'Disposed'. 7. System records the disposal event in the compliance audit log. 8. System updates the government compliance disposal register. 9. If hazardous, system triggers mandatory H&S; notification. 10. System displays disposal confirmation with record ID.
Alternative Flow	A1 – Partial Disposal: Staff specifies partial quantity disposed; system records partial event and updates remaining quantity; status stays 'Active'.
Exception Flow	E1 – Container Not Found (step 3): System shows error; use case ends. E2 – Non-Compliant Method (step 5): System flags method; staff must select approved method. E3 – Already Disposed (step 3): System shows existing record; use case ends.
Business Rules	- Disposal method must be from the approved government list. - Hazardous waste requires a mandatory H&S; report. - Disposal records are immutable once confirmed.
Frequency	Several times per week.
Priority	High — linked directly to compliance audit.

Use Case 4: Generate Compliance Report

Use Case ID	UC-04
Use Case Name	Generate Compliance Report
Actor(s)	H&S; Officer (Primary), Government Regulator (Secondary), System (Tertiary)
Description	The H&S; Officer generates a formal government compliance report on chemical usage and disposal, and optionally submits it to the regulatory portal.

Preconditions	1. H&S; Officer authenticated with appropriate role. 2. Usage and disposal records exist for the requested period. 3. Report templates are configured in the system.
Postconditions	1. Compliance report generated and saved in the compliance archive. 2. If submitted, report transmitted to the government regulatory portal.
Trigger	Upcoming compliance audit, regulatory deadline, or periodic reporting schedule.
Normal Flow	1. H&S; Officer selects 'Generate Compliance Report'. 2. Officer selects report type: Chemical Usage, Disposal, or Combined. 3. Officer specifies date range and department/lab filter. 4. System validates date range and filters. 5. System retrieves all matching usage and disposal records. 6. System compiles data into the government-mandated report template. 7. System generates a formatted PDF report. 8. Officer reviews the report preview. 9. Officer saves the report to the compliance archive. 10. System confirms: 'Report saved successfully.'
Alternative Flow	A1 – Submit to Government Portal: Officer selects 'Submit to Regulatory Portal'; system transmits report via government API; records submission timestamp and confirmation ID. A2 – Schedule Recurring Report: Officer sets monthly/quarterly schedule; system auto-generates and archives reports on schedule.
Exception Flow	E1 – Invalid Date Range (step 4): System shows error; returns to step 3. E2 – No Data Found (step 5): System notifies officer; officer may adjust filters or cancel. E3 – Portal Transmission Failure: System retries up to 3 times; notifies IT Admin if failing.
Business Rules	- Reports must use the government-mandated template format. - Submitted reports are locked post-submission. - All report generation actions are logged in the audit trail.
Frequency	Monthly / quarterly / on-demand.
Priority	Critical

3. Activity Diagrams

Each diagram shows normal flow (blue), exception flows (orange), and alternative flows (green).

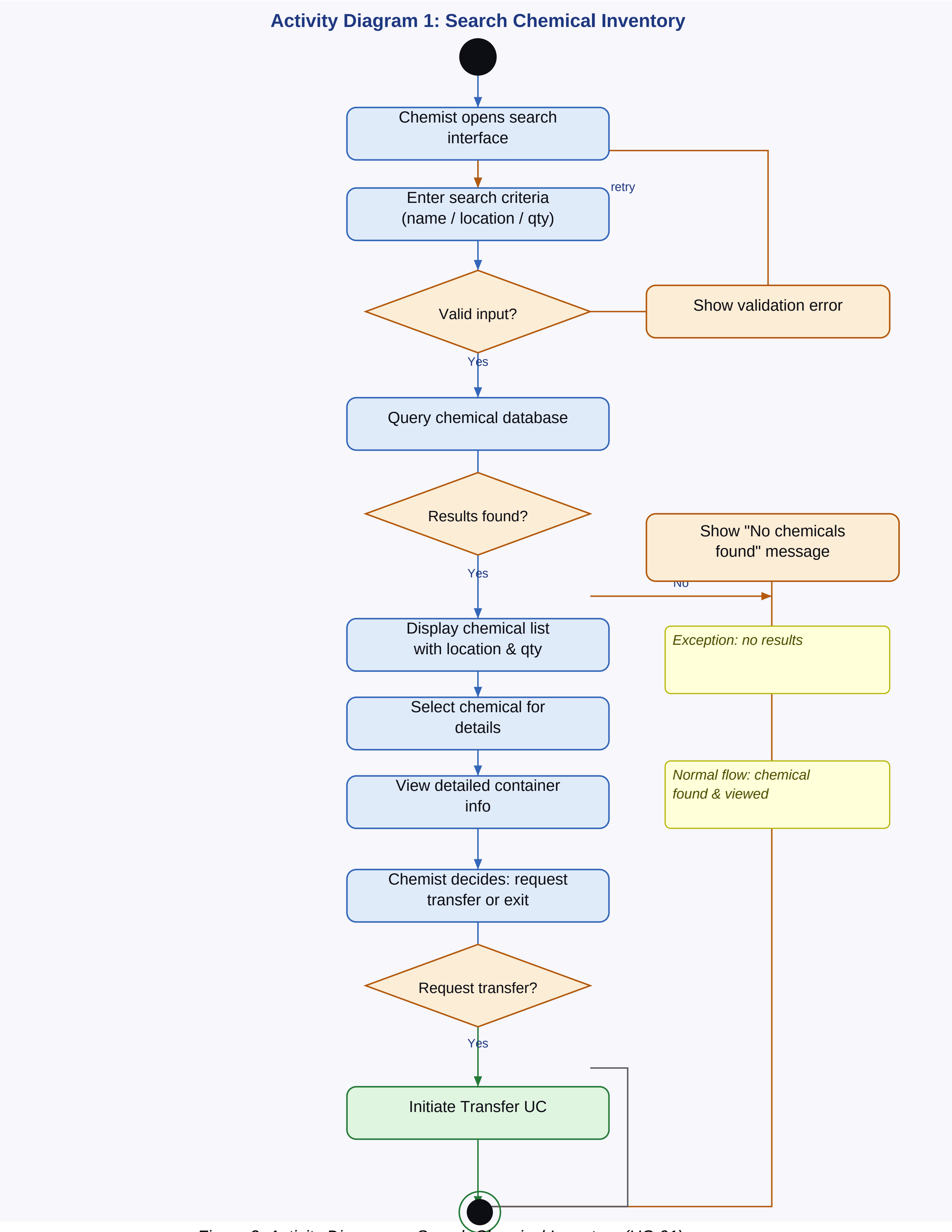

Figure 2: Activity Diagram — Search Chemical Inventory (UC-01)

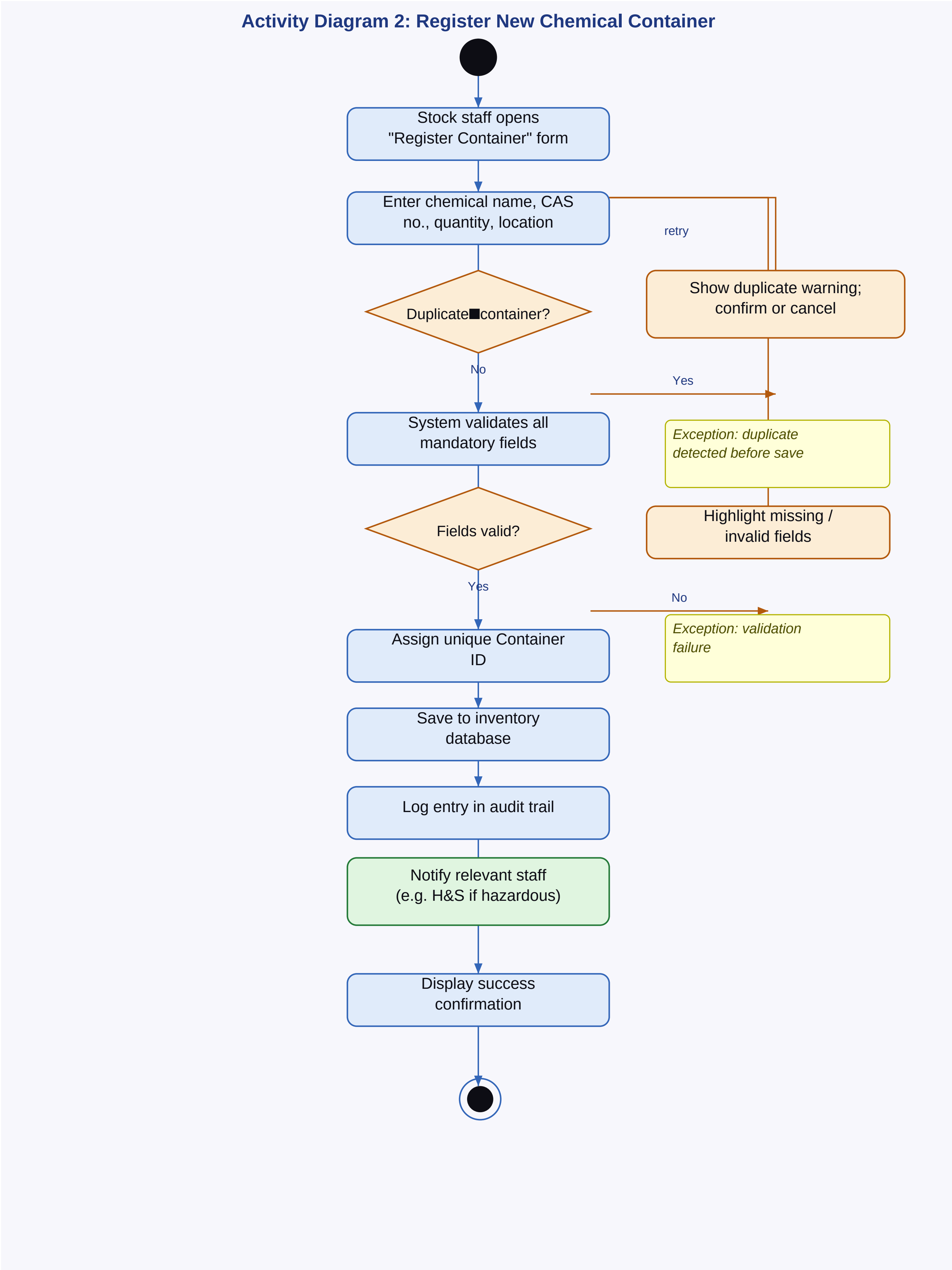

Figure 3: Activity Diagram — Register New Chemical Container (UC-02)

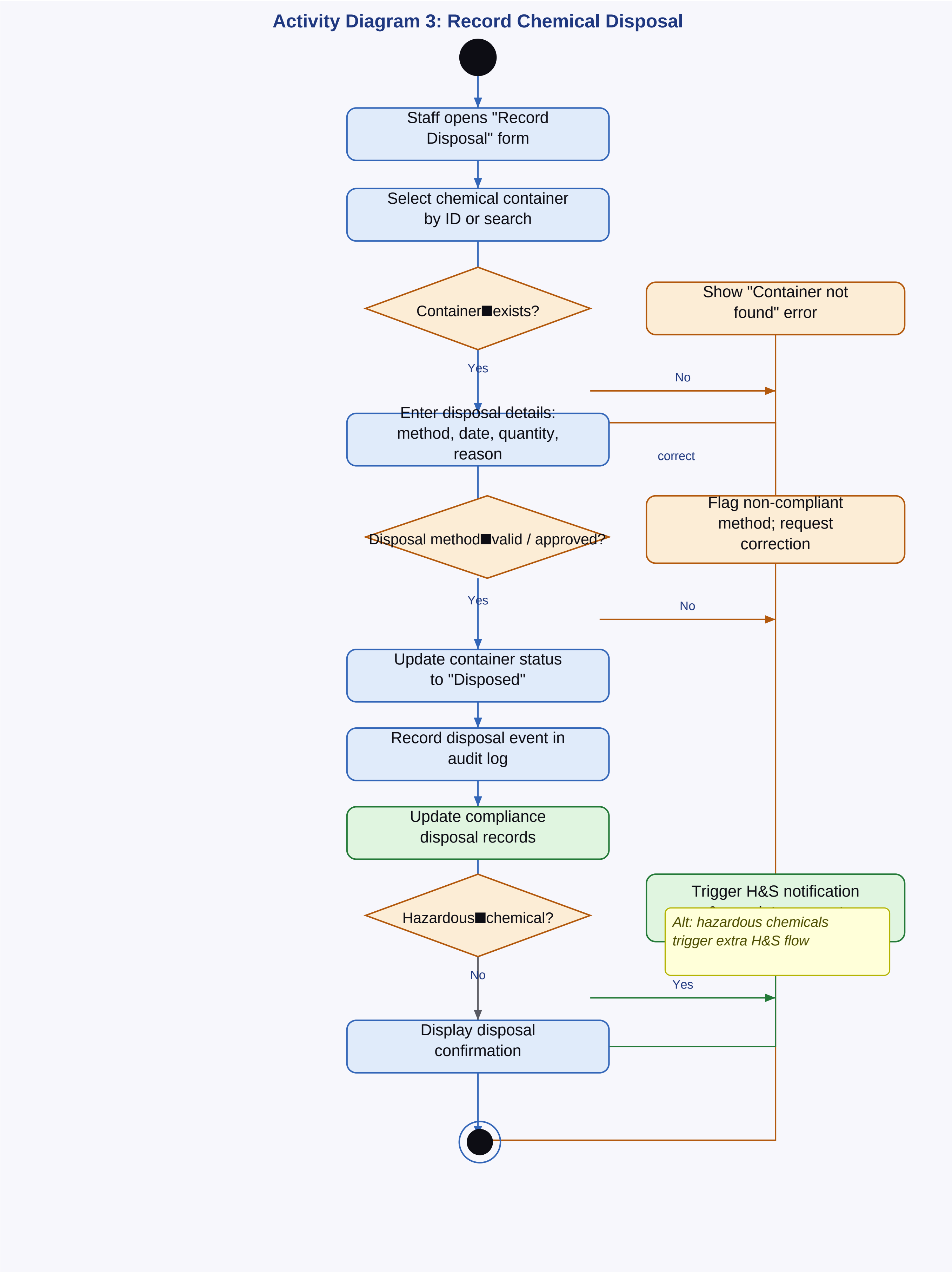

Figure 4: Activity Diagram — Record Chemical Disposal (UC-03)

Activity Diagram 4: Generate Compliance Report

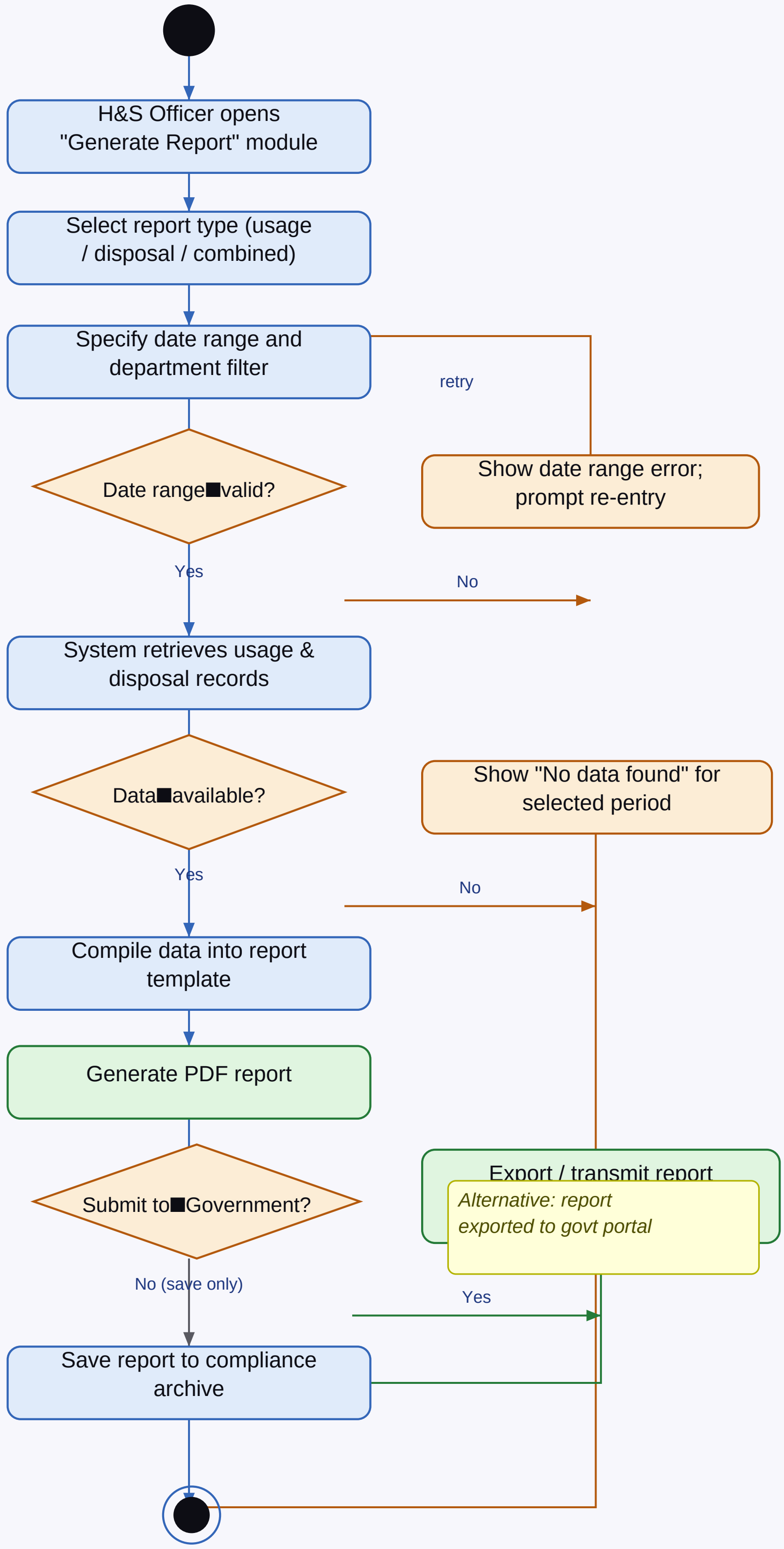

Figure 5: Activity Diagram — Generate Compliance Report (UC-04)